The National Hospital for Neurology and Neurosurgery
Neuroscience Critical Care Unit (NCCU)

University College London Hospitals NHS
NHS Foundation Trust

EXTERNAL VENTRICULAR DRAIN (EVD)

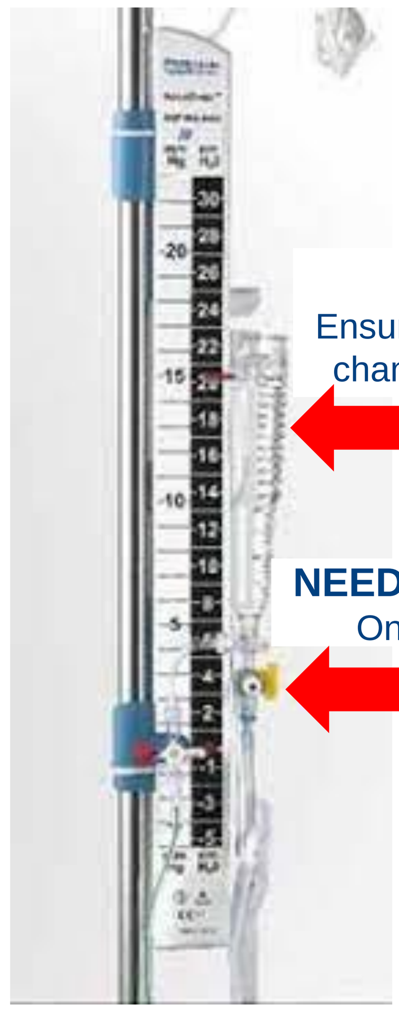
CSF SAMPLING / INTRAVENTRICULAR ANTIBIOTICS /
FLUSHING BLOCKED EVD / LIQUOGUARD EVD: FOLLOW B AND C

A TAKING CSF FROM DRIP CHAMBER

*If CSF sample **ONLY** required take from needle free port on drip chamber*

EQUIPMENT REQUIRED

- Designated **CSF trolley**
  - *Clean with Clinell Universal Wipe, dry well*
  - *Follow with 70% alcohol wipe over entire surface*
- Sterile gloves
- Clinell 2% chlorhexidine in 70% alcohol wipe
- 10ml syringe
- 2 universal sterile containers



DRIP CHAMBER
Ensure nurse knows not to empty chamber prior to CSF sampling

NEEDLE-FREE PORT
On Drip Chamber

SAMPLING PROCEDURE

Take prior to hourly emptying of chamber – nurse will add 6mL sample to hourly drainage total

1. Turn **EVD OFF** *(to prevent syphon effect)*
2. Clean needle free port with Clinell wipe
3. Allow to dry for **30 seconds**
4. Take 6mL CSF from drip chamber
5. Place 3mL CSF in each sterile container
6. Turn **EVD ON**

B TAKING CSF FROM PROXIMAL 3-WAY TAP

PATIENT MUST LIE FLAT FOR PROCEDURE TO REDUCE RISK OF AIR IN SYSTEM

If IT antibiotics also required proximal 3-way port should be used for both taking CSF sample and giving antibiotic

EQUIPMENT REQUIRED

- Designated **CSF trolley** *Clean as above*
- Sterile dressing pack
- Sterile gloves *(or use gloves in dressing pack)* & apron
- Chlorhexidine 0.5% in 70% alcohol
- 2mL + 10mL syringe
- 2 universal sterile containers
- New sterile cap for 3-way tap

SAMPLING PROCEDURE

AN ASSISTANT IS REQUIRED FOR THIS PROCEDURE

Turn EVD OFF before preparing sterile field *to ensure enough CSF in ventricles to prevent collapse when CSF removed (if hourly drainage low EVD may need to be off for longer)*

If LiquoGuard EVD press **PAUSE MODE**

1. Assistant holds EVD line & removes cap from 3-way tap
2. Clean 3-way tap injection port, then lever, then 5cm of EVD line either side of tap
3. Allow to dry for **30 seconds**
4. Place sterile towel from dressing pack on patient's pillow & place EVD on towel
5. Slowly aspirate 2mL CSF & discard
6. Slowly aspirate 6mL CSF - Volume of CSF taken **does not** need to be replaced *(Note that samples <3mL can still be processed)*
7. Place new sterile cap on sampling port
8. Place 3mL CSF in each sterile container
9. Turn **EVD ON**

SENDING the **CSF SAMPLE**

3mL CSF in each universal container for M,C & S / Cell count

0900-1730 - NHNN Specimen Reception
Avoid sending **out of hours** if possible
If essential send via courier &
telephone laboratories to expect sample

C ADMINISTERING INTRAVENTRICULAR (IT) ANTIBIOTICS

Intraventricular antibiotics come in pre-filled sterile syringes labelled for '*INTRAVENTRICULAR USE ONLY*' or for '*INTRATHECAL USE*' & are stored in a designated fridge in NHNN Operations Centre

AMPOULES AND VIALS MUST NOT BE USED

IT gentamicin & vancomycin are 'stock' drugs

*Prescribe to be given between **0800** & **1400** to allow for processing of samples - if 1st dose given 'out of hours' prescribe 2nd dose to be given by 14:00 the following day*

PRIOR TO ADMINISTRATION, IT DRUGS REQUIRE CHECKING BY **TWO** PEOPLE - *ONE OF WHOM MUST BE THE PERSON ADMINISTERING THE DRUG*

EQUIPMENT REQUIRED

As for CSF sampling from 3-way tap

+ Additional equipment

- *Antibiotic in pre-filled sterile syringe for 'INTRAVENTRICULAR USE'*
- *5mL saline flush*
- *0.22 micron filter*
- *Sterile scissors*

CSF SAMPLING + IT ANTIBIOTIC PROCEDURE

1. Take CSF as outlined in SAMPLING PROCEDURE steps B **1-6**
2. SLOWLY inject antibiotic **via filter**
3. SLOWLY flush with saline **via filter** - *aim for CSF removed to = Antibiotic + flush*
4. Leave 3-way tap **OFF TO PATIENT**- *to ensure antibiotic does not drain into system*
5. Place new cap on sampling port
6. Place 3mL CSF in each sterile container
7. Nurse will record time drain turned off
8. Leave **EVD OFF** for **1 hour**
9. Nurse will turn **EVD ON** after **1 hour**

If patient has **BILATERAL EVDs**
Both EVDs should be turned off and FULL DOSE of IT antibiotic administered to each EVD in turn

IF 2 IT ANTIBIOTICS PRESCRIBED
*CSF specimen **only required** prior to **1st antibiotic***

1. *Take CSF specimen*
2. *Give **1st antibiotic***
3. *Leave **EVD OFF** for **1 hour***
4. *Turn **EVD ON** and allow to drain for **1 hour***
5. *Give **2nd antibiotic** – no need to discard CSF prior to giving 2nd antibiotic*
6. *Leave **EVD OFF** for **1 hour***
7. *Turn **EVD ON** after **1 hour***

FLUSHING a BLOCKED EVD

IF EVD NOT DRAINING OR OSCILLATING

A. Check **insertion site -** *Is EVD securing device still in place?*

B. Book **CTH –** *Does scan confirm correct EVD placement?*

If answer is **YES** to A and B above follow flushing procedure

EQUIPMENT REQUIRED

- Designated **CSF trolley**
  Clean with Clinell Universal Wipe, follow with 70% alcohol wipe
- Sterile dressing pack
- Sterile gloves *(or gloves in dressing pack)* & apron
- Chlorhexidine 0.5% in 70% alcohol

+ Additional equipment

- *5mL syringe*
- *5mL saline flush*
- *0.22 micron filter*

FLUSHING PROCEDURE

1. Clean 3-way tap as in Section B 1-4
2. Flush **2mL** saline **via filter** away from patient *– to clear possible debris from sampling port*
3. *Check if EVD now patent*
4. If not, try to **slowly** aspirate maximum **5mL** CSF
5. *Check if EVD now patent*
6. If not, **slowly** flush remaining 3mL saline **via filter** to patient
7. *Check if EVD now patent*
8. If no immediate drainage and patient condition allows, leave for **1 hour** and observe for drainage

Above procedure can be repeated **ONCE**

If drain remains blocked after 2nd attempt escalate to Neurosurgical SpR as *patient may require a new EVD*

If answer is **NO** to A and B above inform Neurosurgical SpR